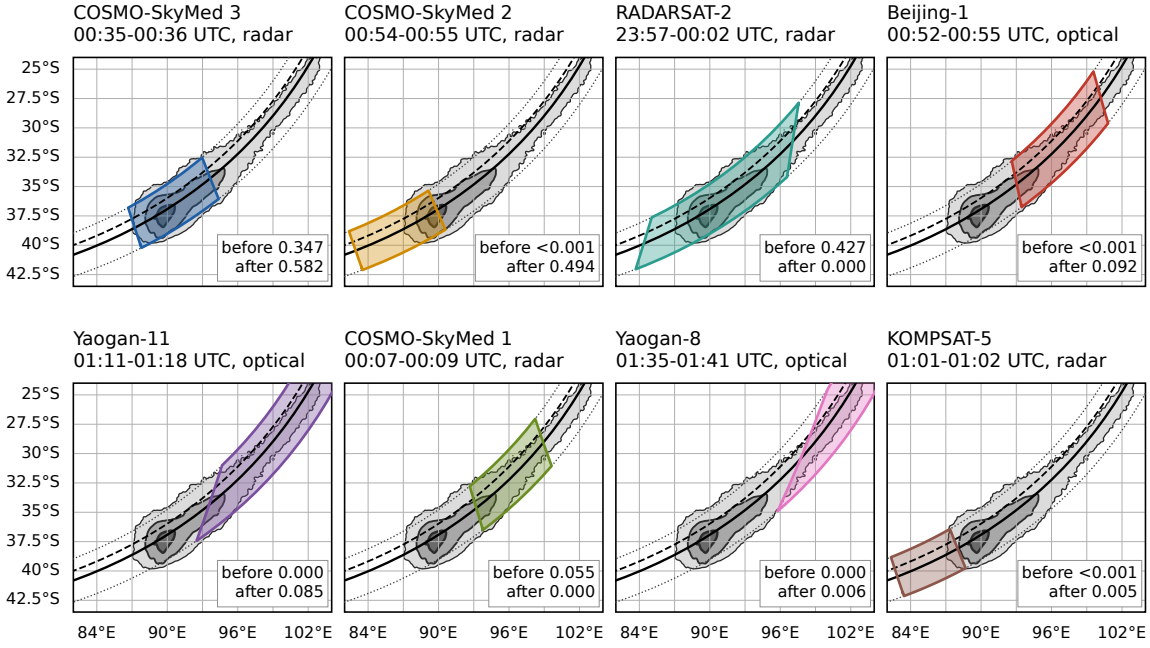
The eight passes that covered the most impact probability, 8 March 2014



(i) Each panel shows one satellite pass. The coloured area is where the sensor could point, cut to 100 NM either side of the 7th arc (solid line; 6th arc dashed). Grey shows where the aircraft probably hit the sea (50, 90, 99 %). 'Before' and 'after' give the share of impact probability in the area up to 30 min before and 0-60 min after impact.
(ii) Impact density: end-of-flight run J, waypoints family (core P = 0.9982), 24 seeds, 1.5 M draws from the rebuilt flight-posterior weights, 00:19 option '00:19 data not used' (D0 total, q = 1/3, other log-on cause silent); 0.05 deg histogram, Gaussian sigma 0.1 deg, highest-density regions. Arcs: shared sea-level arcs. Footprints: SGP4 on March 2014 element sets; SAR band = near-far incidence on the allowed look side(s); optical band = max off-nadir + half swath, sun >= 10 deg at the point. Passes kept if the band covered any impact draw from 30 min before to 60 min after that draw's impact. 'Before' crossings test a fixed point, not the moving aircraft. One SAR scene covers only one beam (about 1-3 % of the band in nominal modes). 19 classified satellites without element sets are not included. KH-11 (USA 186) element set is 210 h from the pass, so its timing is uncertain by minutes.